

*Teaching Notes 8B**

Beyond MM: Asymmetric Information

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Overview

- Focus on fourth MM assumption
 - Symmetric information
- Relaxing this assumption implies “asymmetric information” between the firm and outsiders
 - Firms with private information relative to the market may use capital structure decisions as “signals” which can affect value
- Leads to a more complete view of how firms choose capital structure
 - “Pecking order theory” of capital structure

Objectives

- Understand the “qualitative” aspects that are introduced when we relax MM assumptions
- Understand “quantitatively” what relaxing a particular assumption implies for capital structure

Assumption #5

Symmetric Information

MM Assumptions: Review

- MM assumptions
 - Perfect and complete capital markets
 - No Taxes
 - Bankruptcy is not costly
 - Capital Structure does not affect investment policy and cash-flows
 - Symmetric information
- Now we relax the last assumption
 - There is “asymmetric information”

Using Private Information: Credibility Principle

- Firms can communicate their private information to the market through “signals”
 - But signals have to be convincing (“credible”)
- Signal credible only if costly enough so that the “bad” type (firms which may lie) does not copy
 - **Good type**: firms who have information they want to communicate to the market to improve their value
- Therefore, just sending a press release stating you are the “good” type does not work
 - Called “cheap talk” (easy to copy for everyone)

Credibility Principle and “Different Signals”

Example: Debt as a “Credible Signal”

- Analysts are looking at a semiconductor manufacturer (Letni Inc.) that is developing a new 10nm processor
 - If it works (1/2 chance), Letni's value is \$150B
 - If it fails (1/2 chance), its value is \$70B
- Today's value of Letni Inc. is \$110B
- Can debt be a credible “signal” to convey information that the processor will work or fail?
 - Bankruptcy is prohibitively costly for firm/manager
 - Private information resides with firm/manager

Example: Debt as a “Credible Signal” (contd.)

- Case 1: CEO knows processor doesn't work
 - The “Bad” type of CEO
 - Would he issue debt?
- Potential benefit of issuing more debt than can pay at $t=1$ is “wealth-transfer”
 - Market may think processor will work $\Rightarrow$ firm value increases to 150 from 110
 - Extract money from debt holders
- Potential cost is “bad things” with going bust
 - Bankruptcy costs

Example: Debt as a “Credible Signal” (contd.)

- Case 2: CEO knows processor does work
 - The “good” type of CEO
 - Would he issue debt?
- Potential benefit of issuing debt is the positive signal (the CEO knows the processor works)
 - Market may think processor will work => firm value increases to 150 from 110
- Worry is that “bad type” firm may copy debt issuance making “signal” lose its credibility
 - Pick signal so that bad type can’t possibly copy
 - How much debt is the right/credible signal?

Debt (and other) “Credible Signals”: Bottom line

- Increasing leverage often signals manager’s private positive information
 - Signal is credible if a similar manager with “bad” information would not send it
- Similar signals arise when paying dividends (why?) or repurchasing equity (why?) or when CEO personally owns a substantial fraction of company (why?)
 - Ask “why” the bad type may find it hard to copy
 - These signals also satisfy the “credibility principle”

Issuing Equity: “Adverse Selection”

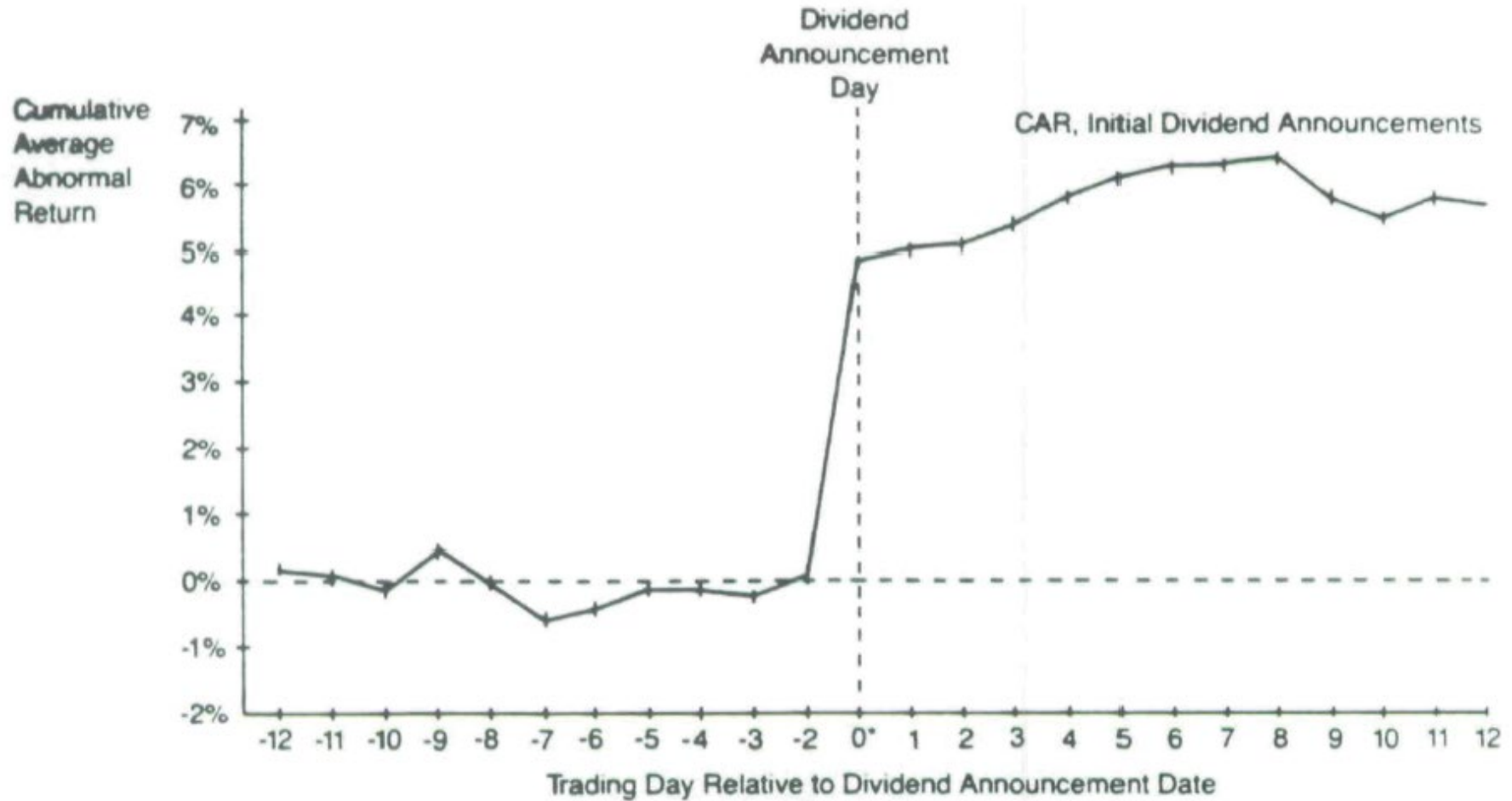
- When firm is overvalued, manager is happy to sell equity (“bad type”)
 - Called “adverse selection”
- When firm is undervalued, manager reluctant to issue new equity (“good type”)
 - May forgo +NPV projects to avoid raising external equity
 - Creates a “cost of external finance” ...why?
 - Issuing shares a negative signal → share price would **drop**
 - Have to issue MORE shares for same investment → **dilution**
 - The cost to existing shareholders from the **drop** and **dilution** could be greater than the positive NPV investment
 - → existing shareholders do not invest by issuing equity

Market Reaction as a Check of “Theory”

Understanding “Real World” Responses

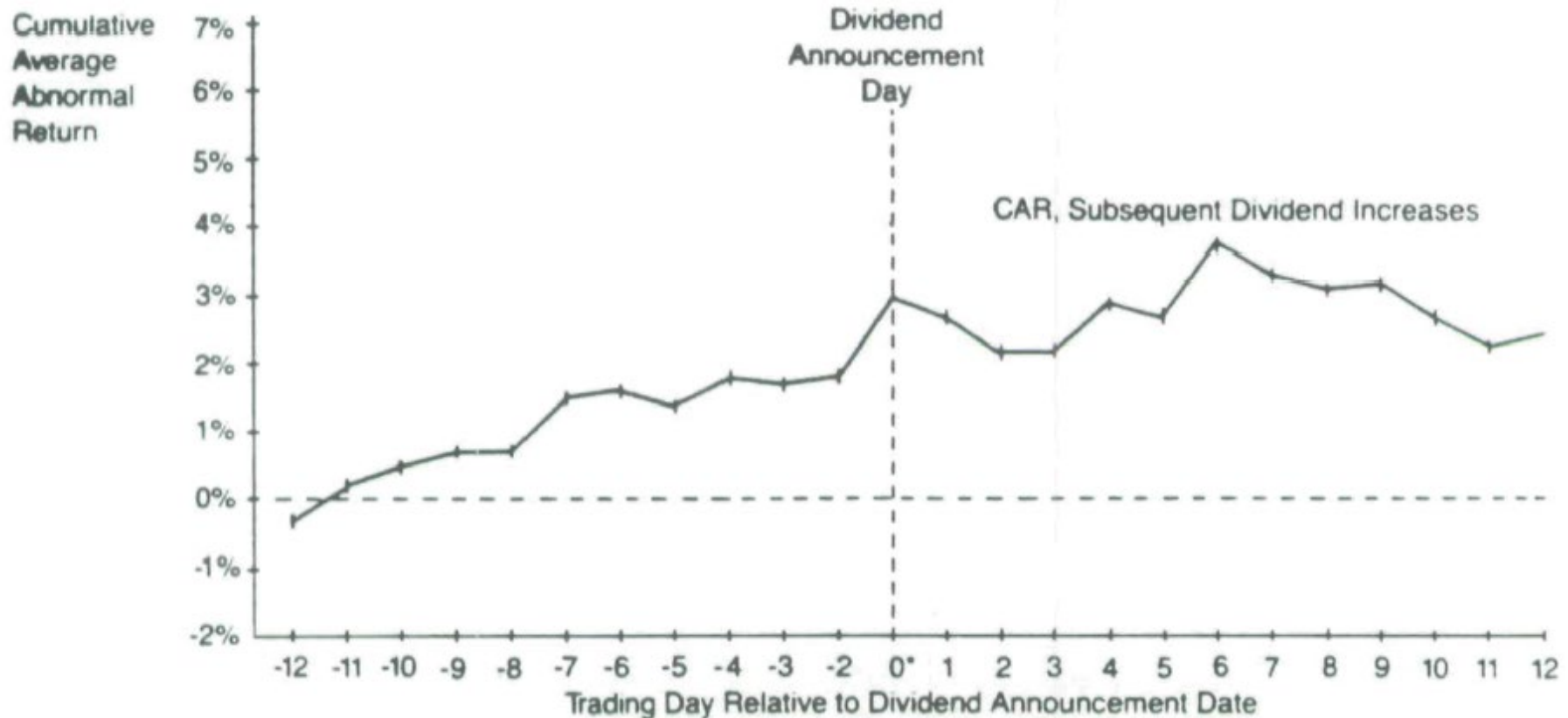
- How does market treat these signals?
 - Examine stock price reactions to announcements of equity transactions
 - Idea is to compare with what “theory” predicts
 - Three types of signals studied
 - (1) Initiation of initial dividends
 - (2) Share repurchases
 - (3) Share issues
- Assess how firms make decisions given these patterns
 - Payouts
 - Security issuance (issue equity, debt, etc)

1. Initiation of Initial Dividends



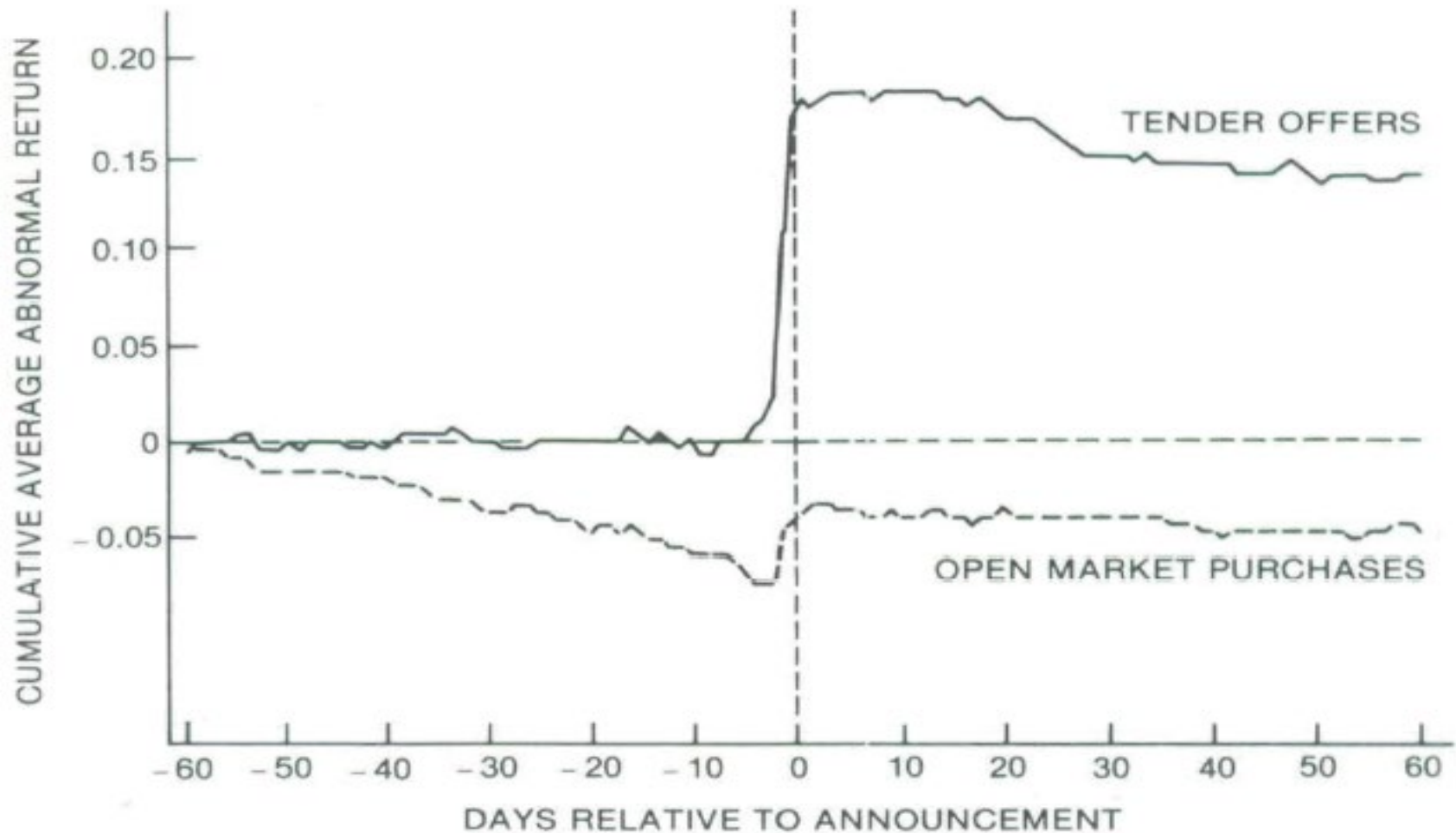
Source: Asquith, Mullins (1986)

Subsequent Dividend Increases



Source: Asquith, Mullins (1986)

2. Share Repurchase



Source: Asquith, Mullins (1986)

Implications for Payout Policy

- Cash at hand can be
 - Kept as buffer against shocks ('inventory management')
 - Reinvested in business
 - Paid out as dividends or used for repurchasing shares
- Considerations for dividends vs. repurchase

1. Signaling

- Dividends signal strong expected future earnings
 - Firm not keeping cash today since it will have enough "internal resources" in the future to finance projects – credible signal?
 - In general, dividend initiation => firm committing to also pay regular future dividends
- Repurchases also seen as a positive signal
 - Signals that managers think firm is undervalued – credible signal?

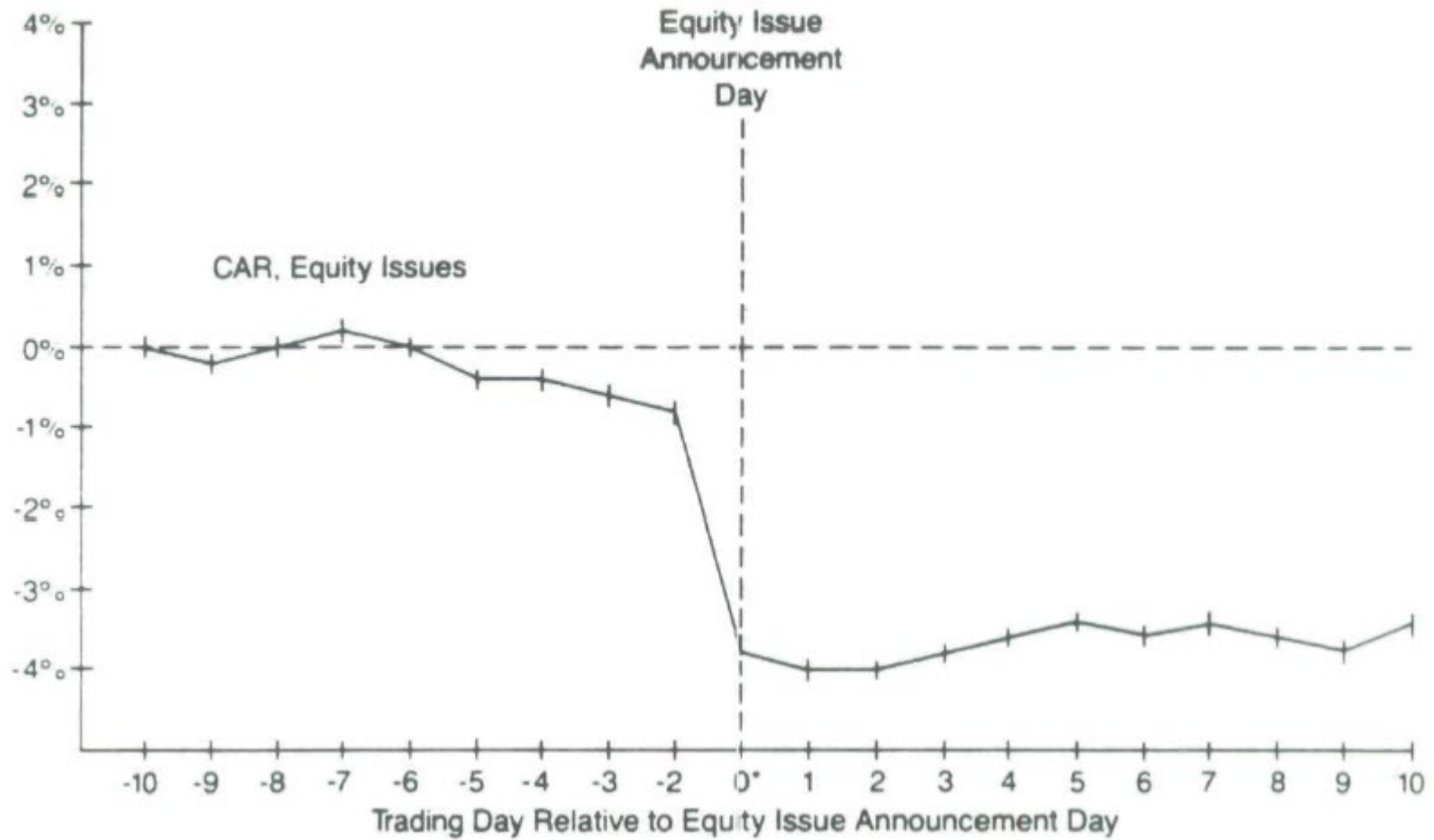
Implications for Payout Policy

- Other considerations for dividends vs. repurchase

2. Taxes

- Dividends are tax disadvantaged (personal taxes) for investors
 - Repurchases → capital gains but can be deferred until investor sells
 - Dividends → taxes that have to be paid immediately
 - Weigh decisions based on investor tax rates
- Overall decision for payout (keep cash vs. reinvest vs. pay dividends/repurchase)
 - A comparison of “NPV calculations”
 - Calculate NPV of keeping buffer – what drives the benefit?
 - Calculate NPV of reinvesting – what drives the benefit?
 - Calculate NPV of paying out – what drives the benefit?

3. Share Issue



Source: Asquith, Mullins (1986)

Issuing Equity

- In MM world, equity issue should leave share price unchanged
- In reality, due to adverse selection, equity issue signals that shares are overvalued
 - Overvalued firms have incentives to issue
- As a result, share price drops after equity issue
 - Average price decrease of 2%
 - Drop can represent up to 30% of the issue capital

Pecking Order Theory

General Principle with Asymmetric Information

- Cost of external finance with a security increases if security sensitive to firm's value, i.e., more sensitive to asymmetric information
- Securities reduce this cost if they are:
 - (1) More senior
 - (2) Shorter maturity
 - (3) Better collateralized
 - (4) Bought by informed providers of capital

Pecking Order Theory of Capital Structure

- General principle followed by firms → Issue securities starting with those that have smallest cost of external finance
 - Called “Pecking Order Theory” of capital structure
- Securities higher in the pecking order are less sensitive to asymmetric information
 - 1) Retained earnings
 - 2) Short term bank debt
 - 3) Public debt
 - 4) Common equity

Pecking Order Theory: Profitable Firms

- Pecking order theory explains why firms with greater profits have less debt
 - Profitable firms need less external financing
 - Can save cost of external finance by not issuing securities (neither equity nor debt)
 - Hence, these firms have higher value and less debt $\Rightarrow$ D/V is lower
 - Recall: Trade-off model could not deliver this prediction

Takeaways: Asymmetric Information

- Managers with private information relative to the market can use capital structure decisions as “signals” to affect firm value
 - + signal: debt, dividends, CEO owns shares...
 - - signal: selling securities
 - Adverse selection makes “selling” a negative signal and creates a cost of external finance → undervalued firms may forgo good investments
- Cost of external finance reduced if information “insensitive” securities are used
 - Gives “Pecking Order” theory of capital structure
 - Explains why profitable firms have low debt